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Electric Vehicle (EV) Market Analysis & Performance Insights

Project details

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1. INTRODUCTION — Overview and Purpose

The project uses Tableau to summarize a multidimensional EV market dataset through nine worksheets, two dashboards and one published story. The aim is to support comparative decisions about sales, battery technology, range, CO2 savings, customer ratings and manufacturing geography.

2. IDEATION PHASE — Problem, Empathy and Brainstorming

Problem: EV market performance is difficult to compare when sales, range, technology and satisfaction are reviewed separately.

Emily Johnson (EV Market Strategy Director): executive category, manufacturer and battery trends. David Chen (Automotive BI Manager): competitive benchmarking, manufacturing geography and satisfaction. Dr. Sarah Williams (Sustainable Mobility Research Analyst): range, chemistry, CO2 savings and sustainable-mobility evidence.

Empathy: stakeholders need quick, trustworthy and interpretable views. Brainstormed options included static reports, individual charts and an integrated Tableau solution; the integrated solution was selected.

3. REQUIREMENT ANALYSIS — Journey, Requirements, DFD and Stack

Journey: connect and inspect data → compare interactive visuals → interpret and share story. Functional requirements: import CSV, create nine analyses, group into two dashboards, publish story. Nonfunctional: clarity, reliability, maintainability and responsible public sharing.

Data flow: EV CSV → Tableau data source → analytical worksheets → dashboards → story → end users. Technology: Tableau Desktop and Tableau Public only; Snowflake and SQL were not implemented.

4. PROJECT DESIGN — Fit, Proposal and Architecture

Problem-solution fit: strategy, BI and research personas need unified comparative market evidence. Proposed solution: nine Tableau charts covering sales, manufacturers, range, CO2, battery capacity, price, chemistry, regional ratings and manufacturing countries.

Architecture: local CSV → Tableau data model and aggregations → nine worksheets → two dashboards → Tableau Public story.

5. PROJECT PLANNING AND SCHEDULING

Project dates: 20–28 October 2026. Retrospective milestone plan: 20–21 data exploration; 22–24 charts; 25–26 dashboards; 27 story and publishing; 28 manual validation and submission. Solo contributor: Gitanjali Panjwani.

6. FUNCTIONAL AND PERFORMANCE TESTING

The student confirmed manual checking of graphs, dashboards and filters. There is no recorded formal timing, stress, concurrency or automated test result. The price-versus-rating chart visibly flags more than 4,000 nulls; interpretations must

acknowledge incomplete records.

7. RESULTS — Nine Visualizations

1. Which EV categories lead annual sales? Sales by Category — horizontal bar chart; two-wheeler scooters lead, with hatchbacks, compact SUVs and sedans also contributing strongly.

2. How do manufacturers compare in sales? Manufacturer Performance — ranked vertical bars for 200 manufacturers; results support competitive benchmarking.

3. How has average driving range changed by model year? Range Trends over Years — line chart for 2018–2026; average range remains near 320–325 km in this dataset.

4. How do CO2 savings differ by drivetrain? CO2 Savings by Drivetrain — AWD averages about 3.9 tons/year, compared with about 3.5–3.7 for FWD and RWD.

5. How does battery capacity relate to range? Battery Capacity vs Range — scatter plot comparing battery capacity (kWh), range (km), and five battery chemistries.

6. Does higher price correspond to customer satisfaction? Price vs Customer Satisfaction — scatter plot by recommended use case; ratings vary at different prices, with >4,000 nulls shown.

7. Which battery technologies dominate the dataset? Battery Tech Market Share — horizontal percent-of-total bars; NMC about 34%, LFP 32%, NCA 17%, Sodium-Ion 10%, Solid-State 7%.

8. How does satisfaction vary across regions? Regional Customer Satisfaction — category-by-region heatmap; displayed mean ratings range approximately 4.06–4.12.

9. How is sales volume distributed by manufacturing country? Manufacturing Country Share — bubble chart; India 159,239,271, China 158,514,269 and Vietnam 157,312,671 annual sales units.

8. ADVANTAGES AND DISADVANTAGES

Advantages: fast comparison, multiple EV dimensions, interactive exploration, public sharing and coherent story.

Disadvantages: dense scatterplots, static data refresh, missing rating values, possible aggregation distortions and no predictive or causal claims.

9. CONCLUSION

Tableau dashboards and story provide a descriptive view of EV market performance and technical tradeoffs. The results help frame questions for strategy, competitive analysis and sustainable-mobility research, while data-quality limitations remain important.

10. FUTURE SCOPE

Improve null handling, create a documented refresh workflow, optimize high-mark scatterplots, introduce governed data access, and evaluate forecasting only after independent validation.

11. APPENDIX — Links and Evidence

Dashboard 1:
https://public.tableau.com/app/profile/gitanjali.panjwani/viz/Dashboard1_17935482618430/Dashboard1?publish=yes

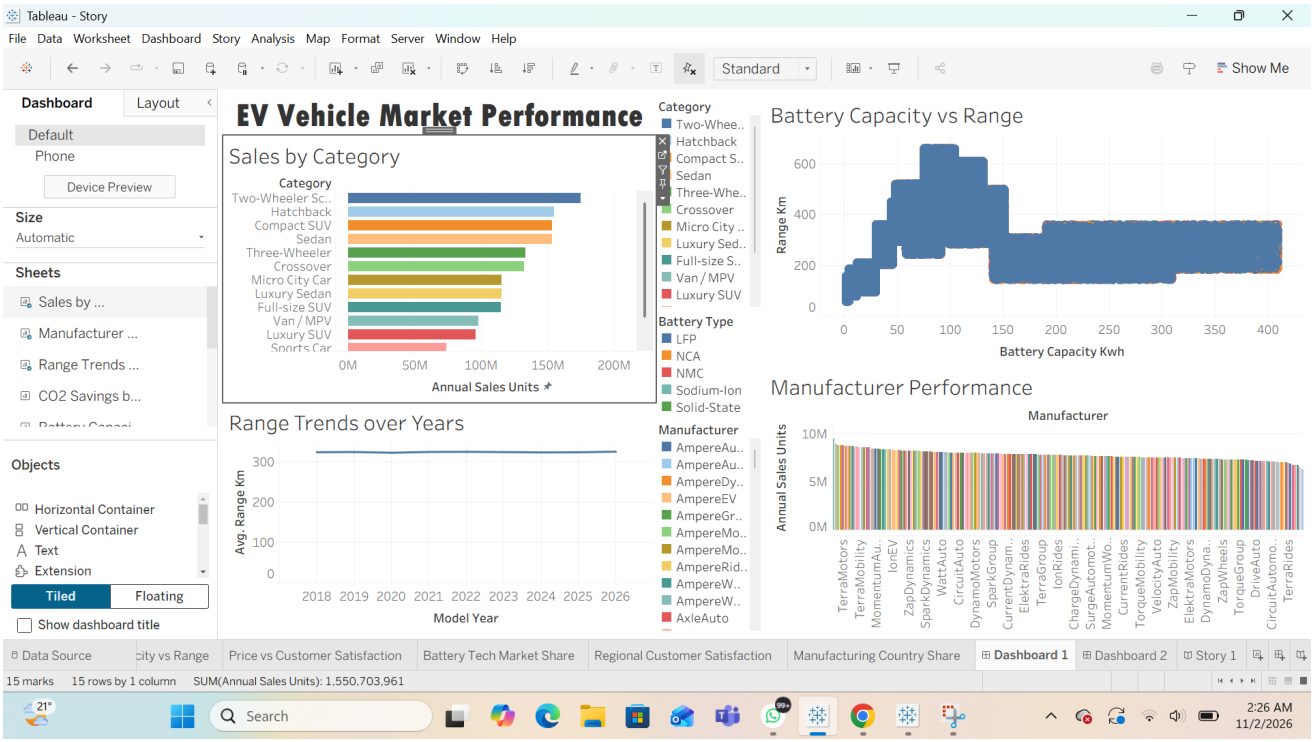
Dashboard 2:
https://public.tableau.com/app/profile/gitanjali.panjwani/viz/Dashboard2_17935484601140/Dashboard2?publish=yes

Story: https://public.tableau.com/app/profile/gitanjali.panjwani/viz/Story_17935476927220/Story1?publish=yes

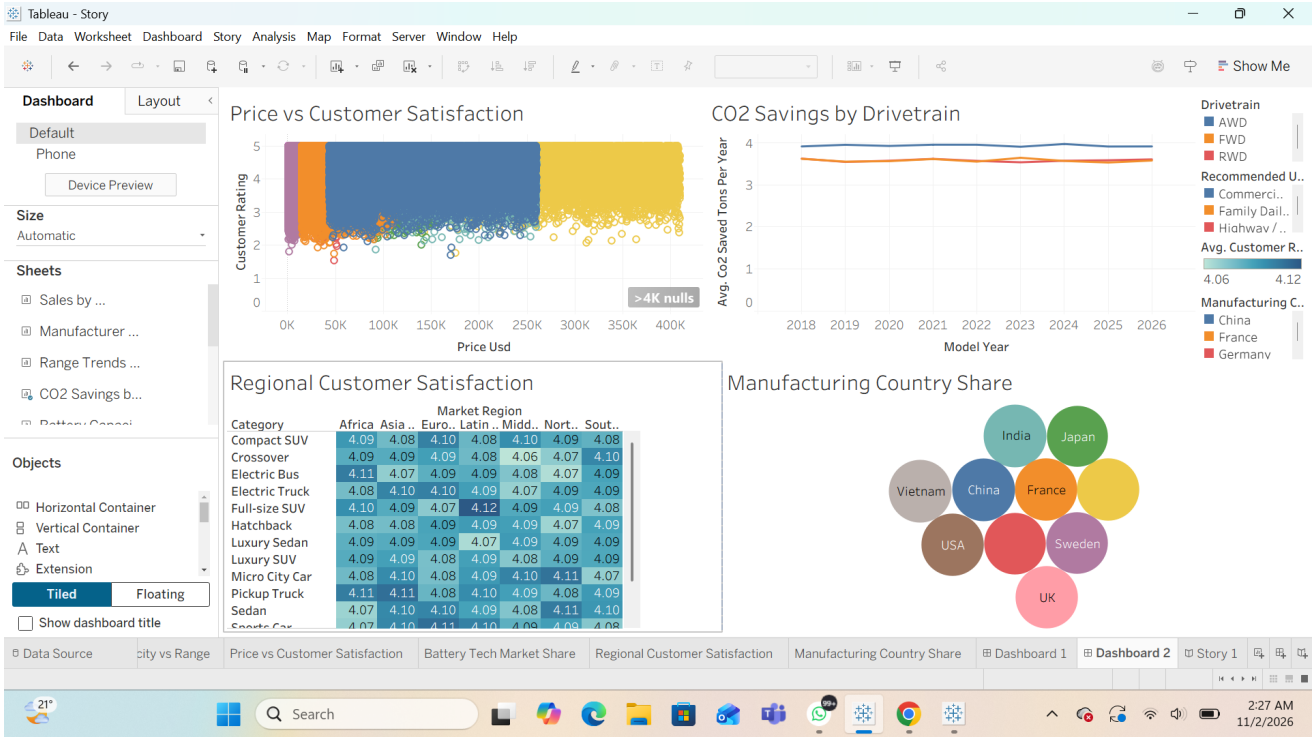
Demonstration video: https://drive.google.com/file/d/1kewhAhWw-mxRimD4bTwy-1InD8BP1cwf/view?usp=drivesdk

Dataset: user-supplied ev_market_train.csv split into .000 and .001 parts. No separate GitHub or source-code repository was provided.

Screenshot evidence — Dashboard 1



Screenshot evidence — Dashboard 2


Tableau - Story
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Show Me
Dashboard
Layout
Default
Phone
Device Preview
Size
Automatic
Sheets
Sales by ...
Manufacturer ...
Range Trends ...
CO2 Savings b...
Objects
Horizontal Container
Vertical Container
Text
Extension
Tiled
Floating
Show dashboard title
Price vs Customer Satisfaction
Customer Rating
Price Usd
>4K nulls
CO2 Savings by Drivetrain
Avg. CO2 Saved Tons Per Year
Model Year
Drivetrain
AWD
FWD
RWD
Recommended U..
Commerci..
Family Dail..
Highway /..
Avg. Customer R..
4.06
4.12
Manufacturing C..
China
France
Germany
Regional Customer Satisfaction
Market Region
Category
Africa
Asia ..
Euro..
Latin ..
Midd..
Nort..
Sout..
Compact SUV 4.09 4.08 4.10 4.08 4.10 4.09 4.08
Crossover 4.09 4.09 4.09 4.08 4.06 4.07 4.10
Electric Bus 4.11 4.07 4.09 4.09 4.08 4.07 4.09
Electric Truck 4.08 4.10 4.10 4.09 4.07 4.09 4.09
Full-size SUV 4.10 4.09 4.07 4.12 4.09 4.09 4.08
Hatchback 4.08 4.08 4.09 4.09 4.09 4.07 4.09
Luxury Sedan 4.09 4.09 4.09 4.07 4.09 4.09 4.09
Luxury SUV 4.09 4.09 4.08 4.09 4.08 4.09 4.09
Micro City Car 4.08 4.10 4.08 4.09 4.10 4.11 4.07
Pickup Truck 4.11 4.11 4.08 4.10 4.09 4.08 4.09
Sedan 4.07 4.10 4.10 4.09 4.08 4.11 4.10
Manufacturing Country Share
India
Japan
Vietnam
China
France
USA
Sweden
UK
Data Source
city vs Range
Price vs Customer Satisfaction
Battery Tech Market Share
Regional Customer Satisfaction
Manufacturing Country Share
Dashboard 1
Dashboard 2
Story 1
Search
2:27 AM
11/2/2026